Kaushik Shroff

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EDUCATION

University of Wisconsin-Madison — Aug 2024 – Dec 2025
Master of Science, Electrical and Computer Engineering — GPA: 3.6/4.0
- **Coursework:** Advanced Computer Architecture, Operating Systems, VLSI Systems Design

Savitribai Phule Pune University — Jul 2020 – Jun 2024
Bachelor of Engineering in Electronics and Telecommunication — GPA: 3.8/4.0
- **Coursework:** Digital Circuits, Microcontrollers, VLSI

WORK EXPERIENCE

Research Assistant — Chiplet Interconnect Design for Mamba LLM Accelerator — May 2025 – Aug 2025
University of Wisconsin-Madison — Madison, WI
Tools: Verilog, SystemVerilog, BookSim, SET, Python
- Integrated **Floret** topology into SET, **-24.6% latency** & **-55.6% DRAM energy** vs. mesh, with a **+139.4% NoC energy** tradeoff.
- Implemented **BF16 Huffman compression** with 32-bit flit packing and dual-phase evaluation (raw vs. compressed).
- Reduced BODY flits by **65%**, cut link transfer latency by **61%**, achieving an overall **2.6x speedup**.
- Contributed to the **DAC 2026 submission "ECLIPS"**, evaluating Huffman-compressed exponent traffic for LLM inference.

PROJECTS

Out Of Order CPU Power Modeling in gem5
Tools: gem5, McPAT, CACTI, Python, C++
- Mapped **20+ micro-architectural components** (e.g., ROB, branch predictor, LSQ) to activation statistics in gem5.
- Extracted **per component energy** using McPAT, improving the modeling accuracy for ALU, caches, and branch units.
- Validated model by **benchmarking 7 workloads**, observing 2× higher power for integer-intensive code.
- Extended gem5 by **adding 5+ custom stat counters** and integrated support for segment-based power analysis.

MUSI Cache Coherence Protocol Implementation in gem5
Tools: gem5, SLICC, C++, Python
- Added **5+ new coherence messages and 10+ state transitions** across cache and directory controllers using SLICC.
- Designed and integrated a **new state of update (U)** to support commutative updates, enabling local aggregation.
- Built **custom C++ micro-benchmarks** to test COUP behavior under conflicting shared-memory access patterns.
- Identified and resolved protocol-level integration issues by instrumenting **fine-grained debug tracing** in cache hierarchy.

RISC-V 5-Stage Pipelined Processor Design
Tools: SystemVerilog, RISC-V ISA
- Implemented a **5-stage pipelined** RISC-V core supporting R-type and Memory instructions.
- Developed a **Hazard Unit** to resolve data dependencies, implementing **operand forwarding**.
- Designed a centralized **control unit** decoding opcode and function fields for ALU and memory control.
- Optimized memory access by engineering a **byte-addressable Data Memory** unit with word-aligned indexing logic.

Graph Neural Network (GNN) Accelerator Design
Tools: SystemVerilog, ModelSim, Design Compiler, Innovus, Virtuoso, PrimeTime, Tcl
- Designed a **GNN accelerator** with MAC, ReLU, and aggregation units for 4-task scheduling on heterogeneous Arm cores.
- Achieved 869 MHz frequency, 3.07 mW power, and **7.91 pJ·ns·mm² EDAP** through optimization like **clock gating & pipelining**.
- Synthesized, placed, and routed using 7nm standard-cell libraries, followed by post-layout **power and timing analysis**.
- **Reduced post-APR power by 5% and pre-APR by 38% using clock gating**, demonstrating hardware-aware optimization.
- Completed **RTL to GDS development**, including verification, synthesis, APR and power/timing closure.

SKILLS

- **Languages:** Python, C++, Bash, Tcl
- **HDL:** Verilog, SystemVerilog
- **Tools:** gem5, McPAT, CACTI, BookSim, SET, ModelSim, Design Compiler, Innovus, Virtuoso, PrimeTime, Git